

Tarea 2: DPS Exercise Set 2 Correlation & Cross-Correlation

Téllez Olmedo Eduardo Iván 321134873

1. Autocorrelation of a discrete signal

Consider the discrete-time signal

$$x[n] = \{1, 2, 3, 2, 1\}, \quad n = 0, \dots, 4$$

- Compute $R_{xx}[k]$ manually.
- Implement the calculation in Python.
- Plot $x[n]$ and $R_{xx}[k]$.
- At which lag is the maximum obtained?
- Describe the symmetry observed in the autocorrelation.

Usando:

$$R_{xx}[k] = \sum_n x[n] x[n-k]$$

$$x[n] = \{1, 2, 3, 2, 1\} \quad n = 0, \dots, 4 \quad N = 5$$

- $K = 0$

$$x[n] \cdot x[n] = 1(1) + 2(2) + 3(3) + 2(2) + 1(1) \\ = 1 + 4 + 9 + 4 + 1 = 19$$

- $K = 1$ $x[n-1] = \{2, 3, 2, 1, 0\}$

$$x[n] \cdot x[n-1] = 1(2) + 2(3) + 3(2) + 2(1) + 1(0) \\ = 2 + 6 + 6 + 2 + 0 = 16$$

- $K = 2$ $x[n-2] = \{3, 2, 1, 0, 0\}$

$$x[n] \cdot x[n-2] = 1(3) + 2(2) + 3(1) + 2(0) + 1(0) \\ = 3 + 4 + 3 + 0 + 0 = 10$$

- $K = 3$ $x[n-3] = \{2, 1, 0, 0, 0\}$

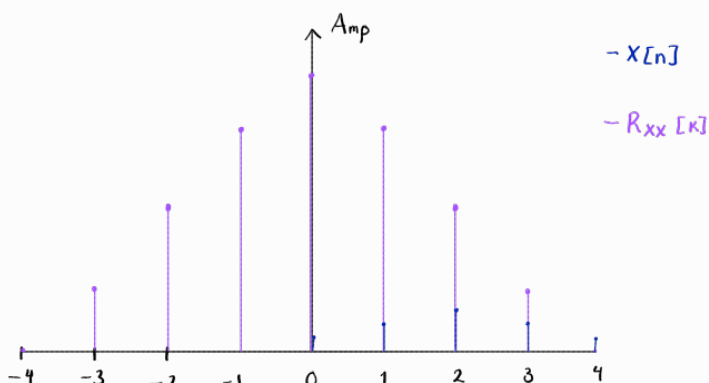
$$x[n] \cdot x[n-3] = 1(2) + 2(1) + 3(0) + 2(0) + 1(0) \\ = 2 + 2 + 0 + 0 + 0 = 4$$

- $K = 4$ $x[n-4] = \{1, 0, 0, 0, 0\}$

$$x[n] \cdot x[n-4] = 1(1) + 2(0) + 3(0) + 2(0) + 1(0) \\ = 1 + 0 + 0 + 0 + 0 = 1$$

Propiedad de Simetría par ($R_{xx}[k] = R_{xx}[-k]$): $R_{xx}[-1] = 16$, $R_{xx}[-2] = 10$, $R_{xx}[-3] = 4$
 $R_{xx}[-4] = 1$

Secuencia obtenida: $R_{xx}[k] = \{1, 4, 10, 16, 19, 16, 10, 4, 1\}$ para $k = -4, \dots, 4$



d) **Lag del máximo:** Ocurre en $k=0$ con valor 19, lo que representa la energía total de la señal $\sum (x[n])^2$

e) **Simetría:** Presenta simetría par respecto al origen $k=0$ ($R_{xx}[k] = R_{xx}[-k]$)

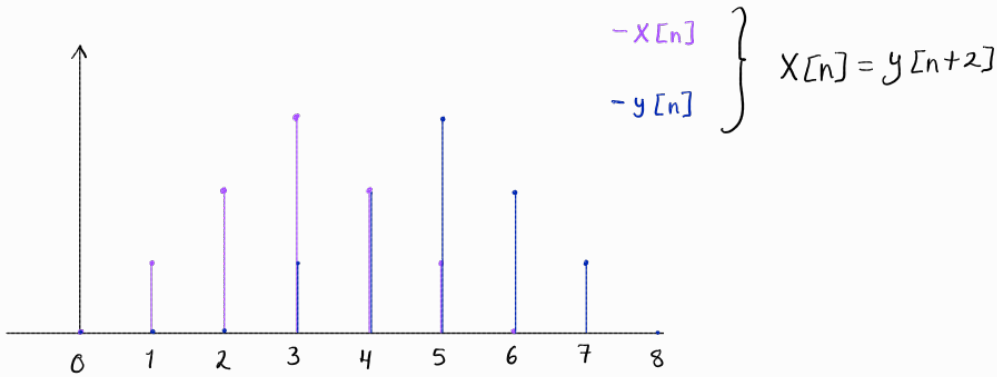
2. Cross-correlation and time displacement

Consider

$$x[n] = \{0, 1, 2, 3, 2, 1, 0\}$$

$$y[n] = \{0, 0, 0, 1, 2, 3, 2, 1, 0\}$$

- Plot both signals.
- Compute $R_{xy}[k]$.
- Plot the cross-correlation.
- Determine the lag corresponding to its maximum.
- Explain what this lag tells you about the relative position of the signals.



$$K = -2 \quad y[n+2] = \{0, 1, 2, 3, 2, 1, 0\}$$

$$X[n] \cdot y[n+2] = 0(0) + 1(1) + 2(2) + 3(3) + 2(2) + 1(1) + 0(0) = 0 + 1 + 4 + 9 + 4 + 1 + 0 = 19$$

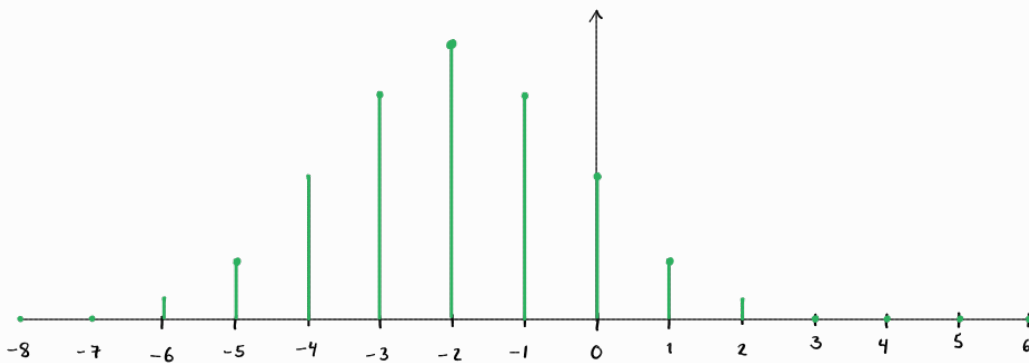
$$K = -1 \quad y[n+1] = \{0, 0, 1, 2, 3, 2, 1, 0\} \quad x[n] = \{0, 1, 2, 3, 2, 1, 0, 0\}$$

$$X[n] \cdot y[n+1] = 0(0) + 1(0) + 2(1) + 3(2) + 2(3) + 1(2) + 0(1) + 0(0) = 0 + 0 + 2 + 6 + 6 + 2 + 0 + 0 = 16 = R_{xy}[-3]$$

$$K = 0 \quad y[n] = \{0, 0, 0, 1, 2, 3, 2, 1, 0\} \quad x[n] = \{0, 1, 2, 3, 2, 1, 0, 0, 0\}$$

$$X[n] \cdot y[n] = 0(0) + 1(0) + 2(0) + 3(1) + 2(2) + 1(3) + 0(2) + 0(1) + 0(0) = 10 = R_{xy}[-4]$$

$$X[n] \cdot y[n-1] = 4 = R_{xy}[-5], \quad X[n] \cdot y[n-2] = 1 = R_{xy}[-6]$$



d) **Lag del máximo:** Se obtiene en $K = -2$ con un valor máximo de 19

e) **Relación entre señales:** Se observa que la señal y es igual a la señal x retrasada 2 muestras
 $y[n] = x[n-2]$

3. Detecting a known pattern

A receiver observes

$$x[n] = \{0, 0, 1, -1, 2, 1, -1, 0, 0\}$$

and searches for the pattern

$$p[n] = \{1, -1, 2, 1, -1\}$$

- Compute the cross-correlation between $x[n]$ and $p[n]$.
- Plot the result.
- Determine where the pattern occurs in $x[n]$.
- Explain why correlation can be used as a detection mechanism.

Para $n=0$ ($x[0, \dots, 4] = \{0, 0, 1, -1, 2\}$):

$$x[n] \cdot p[n] = 0(1) + 0(-1) + 1(2) + (-1)(1) + 2(-1) = 2 - 1 - 2 = -1$$

Para $n=1$ ($x[1, \dots, 5] = \{0, 1, -1, 2, 1\}$):

$$x[n] \cdot p[n] = 0(1) + 1(-1) + (-1)(2) + 2(1) + 1(-1) = -1 - 2 + 2 - 1 = -2$$

Para $n=2$ ($x[2, \dots, 6] = \{1, -1, 2, 1, -1\}$):

$$x[n] \cdot p[n] = 1(1) + (-1)(-1) + 2(2) + 1(1) + (-1)(-1) = 1 + 1 + 4 + 1 + 1 = 8$$

Para $n=3$ ($x[3, \dots, 7] = \{-1, 2, 1, -1, 0\}$):

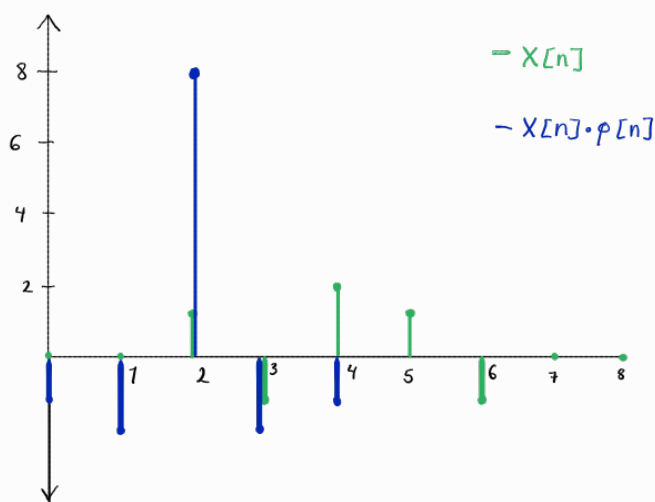
$$x[n] \cdot p[n] = -1(1) + 2(-1) + 1(2) + (-1)(1) + 0(-1) = -1 - 2 + 2 - 1 - 0 = -2$$

Para $n=4$ ($x[4, \dots, 8] = \{2, 1, -1, 0, 0\}$):

$$x[n] \cdot p[n] = 2(1) + 1(-1) + (-1)(2) + 0(1) + 0(-1) = 2 - 1 - 2 + 0 - 0 = -1$$

El pico máximo ocurre al iniciar en $n=2$

La correlación actúa como detector porque el producto punto alcanza su valor máximo cuando la forma del patrón y la señal observada coinciden exactamente en fase y forma.



4. Autocorrelation of a sinusoid

Generate

$$x(t) = \sin(2\pi \cdot 5t), \quad 0 \leq t \leq 2 \text{ s}, \quad f_s = 100 \text{ Hz}$$

- Plot the signal.
- Compute and plot its autocorrelation.
- Measure the distance, in samples, between consecutive autocorrelation maxima.
- Convert this distance into seconds.
- What relationship do you observe between this value and the frequency of the original signal?

a)



c)

Periodo de la señal (T_0)

$$T_0 = \frac{1}{f_0} = \frac{1}{5 [\text{Hz}]} = 0.2 [\text{s}]$$

Periodo de muestreo (T_s)

$$T_s = \frac{1}{f_s} = \frac{1}{100 [\text{Hz}]} = 0.01 [\text{s}]$$

Cálculo de número de muestras

$$N = \frac{T_0}{T_s} = \frac{0.2}{0.01} = 20 \text{ muestras}$$

Los máximos adyacentes se encontrarán en múltiplos enteros de N ($\pm 20, \pm 40, \pm 60$, etc.)

d) Conversión a segundos

$$\Delta t = \frac{N}{f_s} = \frac{20}{100 [\text{Hz}]} = 0.2 [\text{s}]$$

e) Relación entre Δt y f_0

$\Delta t = f_0 \therefore$ La correlación preserva la periodicidad de la señal y se puede estimar su frecuencia sin utilizar el dominio de la frecuencia

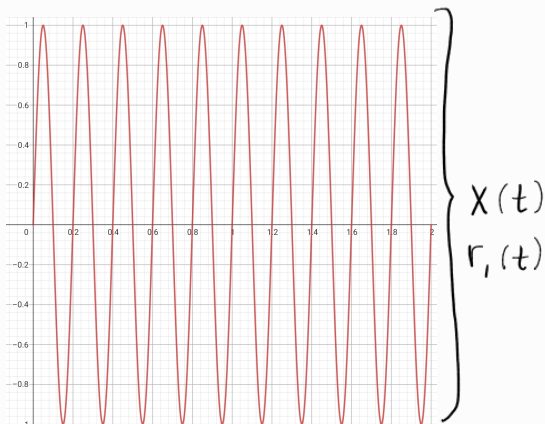
5. Comparing signals with different frequencies

Generate

$$x(t) = \sin(2\pi \cdot 5t)$$

$$r_1(t) = \sin(2\pi \cdot 5t), \quad r_2(t) = \sin(2\pi \cdot 8t)$$

- Plot all three signals.
- Calculate the cross-correlation between $x(t)$ and each reference.
- Compare the maximum absolute correlation obtained in each case.
- Which reference provides the strongest response? Explain why.

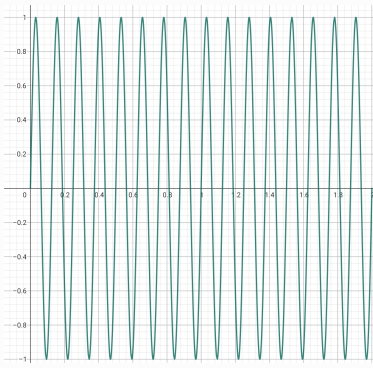


c) Correlaciones de $x(t)$

$$x(t) = r_1(t) \therefore R_{xr_1} = 1$$

$$x(t) \neq r_2(t) \therefore R_{xr_2} \approx 0.52$$

$$\therefore R_{xr_1} > R_{xr_2}$$



d) Significado de referencias

La referencia con $r_1(t)$ es más fuerte ya que $x(t)$ y $r_1(t)$ tienen la misma frecuencia. Mientras que la referencia con $r_2(t)$ es menor ya que NO comparten frecuencias e implica que sus máximos se localizan en puntos diferentes

6. Finding a hidden frequency

The observed signal is

$$x(t) = 2 \sin(2\pi \cdot 7t)$$

Assume its frequency is one of $\{3, 5, 7, 9, 11\}$ Hz. For every candidate frequency f , generate

$$r_f(t) = \sin(2\pi ft)$$

- Compute the correlation of $x(t)$ with each reference signal.
- Store the maximum absolute correlation obtained for each frequency.
- Plot correlation magnitude versus candidate frequency.
- Determine the frequency present in $x(t)$.

d) Frecuencia $r_f(t)$

Como vimos anteriormente, si las frecuencias coinciden se presentará la correlación máxima, por lo tanto la frecuencia de $r_f(t)$ es 7 [Hz]

7. Detecting two frequencies

Consider

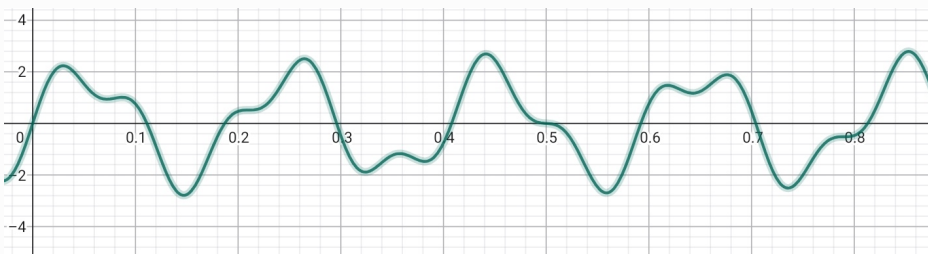
$$x(t) = 2 \sin(2\pi \cdot 5t) + 0.8 \sin(2\pi \cdot 12t)$$

Sample the signal at $f_s = 100 \text{ Hz}$ for 2 seconds. Test candidate frequencies $f = \{2, 3, 4, \dots, 20\} \text{ Hz}$ using

$$r_f(t) = \sin(2\pi ft)$$

- Plot $x(t)$.
- Compute its correlation with every reference sinusoid.
- For each frequency, obtain a single measure representing the strength of the correlation.
- Plot correlation magnitude versus frequency.
- Identify the two dominant frequencies.
- Can the relative amplitudes of the two components be inferred from the correlation results? Explain.

a)



e) Frecuencias dominantes

Las frecuencias dominantes son 5 [Hz] y 12 [Hz] , ya que son las frecuencias contenidas en $x(t)$.

f) Amplitud relativa

$$A = \frac{\max |R_{x r_{f1}}|}{\max |R_{x r_{f2}}|} = \frac{2}{0.2} = 10 \therefore$$

El pico de 5 [Hz] (r_{f1}) es 10 veces mayor que el pico de 12 [Hz]

8. Phase and sinusoidal detection

Consider two signals with the same frequency but different phase:

$$x_1(t) = \sin(2\pi \cdot 8t)$$

$$x_2(t) = \sin(2\pi \cdot 8t + \pi/3)$$

Initially use the sine reference

$$r_s(t) = \sin(2\pi \cdot 8t)$$

a) Plot $x_1(t)$, $x_2(t)$, and $r_s(t)$.

b) Compute the correlation between $x_1(t)$ and $r_s(t)$.

c) Compute the correlation between $x_2(t)$ and $r_s(t)$.

d) Compare the two results. Both signals have the same frequency. Why are their correlation values different?

e) Now use the cosine reference $r_c(t) = \cos(2\pi \cdot 8t)$ and compute its correlation with $x_1(t)$ and $x_2(t)$.

f) Considering the responses obtained using both sine and cosine references, explain why both are useful when the signal phase is unknown.

d) Comparación entre fases y correlación

$$\sum_n \sin(\omega n) \sin(\omega n + \phi) \approx \frac{N}{2} \cos \phi \quad \leftarrow \text{Senoidales de igual frecuencia}$$

Para x_1 ($\phi=0$) $\therefore \cos(0)=1$, la correlación es máxima

Para x_2 ($\phi=\frac{\pi}{3}$) $\therefore \cos(\frac{\pi}{3})=0.5$, la correlación disminuye

f) Utilidad ante fases desconocidas

Porque usar una sola referencia puede fallar

$$M = (C_{\text{seno}})^2 + (C_{\text{coseno}})^2$$

$$x_1: \sqrt{(100)^2 + (0)^2} = 100 \quad x_2: \sqrt{(50)^2 + (86.6)^2} = 100$$

Al tener la misma amplitud y frecuencia ambas señales producirán una magnitud idéntica
La información queda contenida en la relación entre ambos componentes

$$\phi = \arctan\left(\frac{C_{\text{coseno}}}{C_{\text{seno}}}\right)$$

9. Frequency detection in white Gaussian noise

Generate the sampled signal

$$x[n] = \sin(2\pi \cdot 6n/f_s) + 0.7 \sin(2\pi \cdot 15n/f_s) + w[n]$$

using $f_s = 100$ Hz. Here $w[n]$ denotes zero-mean white Gaussian noise: independent Gaussian samples with mean 0 and a selected standard deviation σ .

a) Plot the clean signal and the noisy signal.

b) Test reference frequencies from 1 to 20 Hz.

c) Calculate the correlation response for every candidate frequency.

d) Plot correlation magnitude versus frequency.

e) Determine whether the frequencies 6 Hz and 15 Hz can still be detected.

f) Repeat for at least three different values of σ and discuss what happens as the noise level increases.

e) Frecuencias detectables

Ambas aún son detectables bajo niveles moderados de σ

El pico de la frecuencia de 6 [Hz] (1) es mayor al pico de la frecuencia de 15 [Hz] (0.7)

f) Incremento de ruido

Cuando aumenta σ el "piso de ruido" de correlación sube y los picos de 6 [Hz] y 15 [Hz] aún son detectables pero a σ extremadamente altos (ejemplo $\sigma > 10$), los picos aleatorios de ruido pueden superar los picos de las señales reales causando falsas detecciones

10. Build a frequency detector

Design a Python function:

```
detect_frequencies(x, fs, frequencies)
```

The function must receive a sampled signal x , sampling frequency fs , and a list of candidate frequencies, and return a measure of the presence of every candidate frequency.

Test it using

$$x(t) = 1.5 \sin(2\pi \cdot 4t) + 0.7 \sin(2\pi \cdot 9t + \pi/4) + 2 \sin(2\pi \cdot 16t - \pi/3)$$

- Search frequencies from 1 to 20 Hz.
- Plot the original signal.
- Plot detector response versus frequency.
- Identify the three dominant frequencies.
- Rank them according to their detected strength.
- Explain why a detector using both sine and cosine references is preferable to one using only sine.

d) Frecuencias dominantes

Las frecuencias detectadas aproximadamente son: 4 [Hz], 9 [Hz] y 16 [Hz]

e) Ranking de fuerzas

- 16 Hz (amplitud original = 2)
- 4 Hz (amplitud original = 1.5)
- 9 Hz (amplitud original = 0.7)

f) Uso de referencias seno y coseno

* Problema al usar solo seno: Si la frecuencia coincide pero la señal está desfasada 90° (actúa como coseno), la correlación es cero y el detector falla

* Ventaja de usar seno + coseno: Al combinar las referencias ortogonales usando $\sqrt{\sin^2 + \cos^2}$, se aplica la identidad trigonométrica $\sin^2(\theta) + \cos^2(\theta) = 1$, haciendo al detector inmune a los desfases de la señal